
Evidencias de ejecucion

Microservicios y resiliencia en la nube con Spring Cloud

ESTUDIANTE

Jean Luc Santoni Camilo

ASIGNATURA

Desarrollo Backend III — PBY2203

ACTIVIDAD

Experiencia 3, Semana 8

PROYECTO

Modernizacion del sistema legacy del Banco XYZ

REPOSITORIO

github.com/JeanSantoni02/backend3

1. Que se implemento

El proyecto parte de los procesos batch del Banco XYZ y los lleva a una arquitectura de microservicios segura, resiliente y orientada a eventos. Esta entrega cubre los tres puntos pedidos para la semana 8 sobre la base construida en las semanas 6 y 7.

Requerimiento	Como se resolvio
OAuth2.0	Servidor de autorizacion propio que emite JWT firmados con RSA mediante el flujo <i>client_credentials</i> . El servicio de dominio actua como resource server y autoriza por scope: solo el BFF de cajeros puede retirar.
Imagenes Docker	Ocho Dockerfile en dos etapas. Maven compila en la primera y la segunda copia solo el jar sobre un JRE, con un usuario sin privilegios.
docker-compose	Orquesta diez componentes con healthcheck y <i>condition: service_healthy</i> , de modo que ningun microservicio arranca antes que su infraestructura.

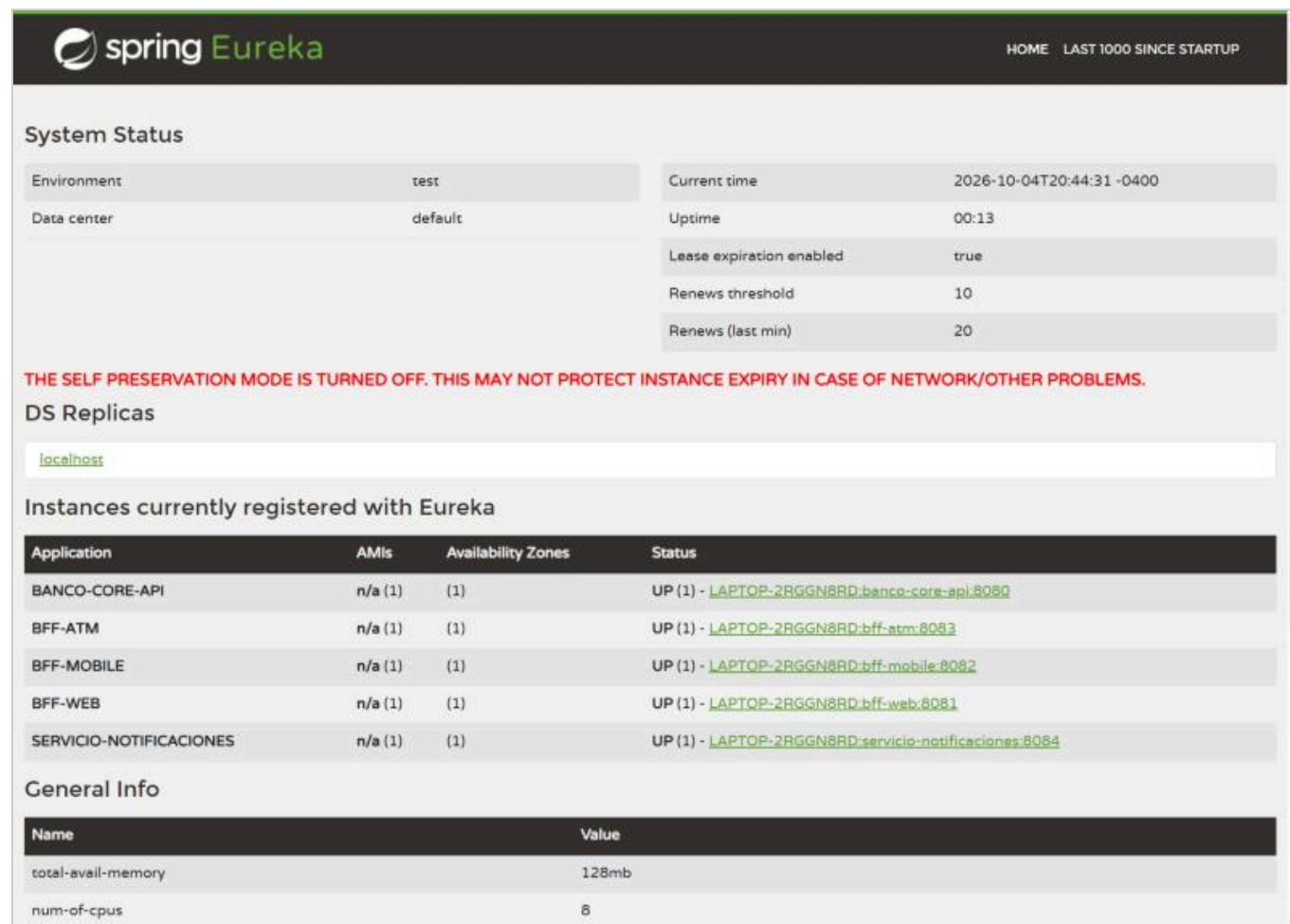
Modulo	Puerto	Funcion
config-server	8888	Configuracion centralizada
eureka-server	8761	Service Discovery
auth-server	9000	Servidor de autorizacion OAuth2
banco-core-api	8080	Dominio. Unico con acceso a PostgreSQL
servicio-notificaciones	8084	Consumidor de eventos de transaccion
bff-web	8081	BFF del portal
bff-mobile	8082	BFF de la app movil
bff-atm	8083	BFF de cajeros, con tolerancia a fallos
batch-migration	-	Carga inicial de los datos (semana 3)

2. Capturas de la plataforma en ejecucion

Los ocho servicios levantados en la maquina local, con los datos cargados por el batch en PostgreSQL.

Service Discovery

Los cinco microservicios registrados en Eureka y en estado UP. El aviso rojo confirma que la autopreservacion esta desactivada, de modo que una instancia caida desaparece del registro en vez de quedar colgada.



The screenshot displays the Spring Eureka web interface. At the top, the 'spring Eureka' logo is on the left, and 'HOME LAST 1000 SINCE STARTUP' is on the right. The 'System Status' section shows environment details (test, default) and system metrics (Current time: 2026-10-04T20:44:31 -0400, Uptime: 00:13, Lease expiration enabled: true, Renew threshold: 10, Renew last min: 20). A red warning message states: 'THE SELF PRESERVATION MODE IS TURNED OFF. THIS MAY NOT PROTECT INSTANCE EXPIRY IN CASE OF NETWORK/OTHER PROBLEMS.' Below this, the 'DS Replicas' section shows 'localhost'. The 'Instances currently registered with Eureka' table lists five applications: BANCO-CORE-API, BFF-ATM, BFF-MOBILE, BFF-WEB, and SERVICIO-NOTIFICACIONES, all with status 'UP (1)'. The 'General Info' section shows system metrics: total-avail-memory (128mb) and num-of-cpus (8).

System Status			
Environment	test	Current time	2026-10-04T20:44:31 -0400
Data center	default	Uptime	00:13
		Lease expiration enabled	true
		Renews threshold	10
		Renews (last min)	20

THE SELF PRESERVATION MODE IS TURNED OFF. THIS MAY NOT PROTECT INSTANCE EXPIRY IN CASE OF NETWORK/OTHER PROBLEMS.

DS Replicas

localhost

Instances currently registered with Eureka			
Application	AMIs	Availability Zones	Status
BANCO-CORE-API	n/a (1)	(1)	UP (1) - LAPTOP-2RGGN8RD-banco-core-api:8080
BFF-ATM	n/a (1)	(1)	UP (1) - LAPTOP-2RGGN8RD-bff-atm:8083
BFF-MOBILE	n/a (1)	(1)	UP (1) - LAPTOP-2RGGN8RD-bff-mobile:8082
BFF-WEB	n/a (1)	(1)	UP (1) - LAPTOP-2RGGN8RD-bff-web:8081
SERVICIO-NOTIFICACIONES	n/a (1)	(1)	UP (1) - LAPTOP-2RGGN8RD-servicio-notificaciones:8084

General Info	
Name	Value
total-avail-memory	128mb
num-of-cpus	8

Configuracion centralizada

El config server respondiendo la configuracion de bff-atm. Entrega dos fuentes: bff-atm.yml con los umbrales del circuito y los reintentos, y application.yml con lo comun a todos los servicios. Nada de esto vive dentro del jar del BFF.

```
pretty-print
{
  "name": "bff-atm",
  "profiles": [
    "default"
  ],
  "label": null,
  "version": null,
  "state": null,
  "propertySources": [
    {
      "name": "classpath:/config-repo/bff-atm.yml",
      "source": {
        "banco.core-api.url": "${CORE_API_URL:http://localhost:8080}",
        "banco.ate.monto-minimo": 1000,
        "banco.ate.monto-maximo": 200000,
        "banco.ate.denominacion": 1000,
        "resilience4j.circuitbreaker.circuit-breaker-aspect-order": 3,
        "resilience4j.circuitbreaker.instances.coreApi.sliding-window-type": "COUNT_BASED",
        "resilience4j.circuitbreaker.instances.coreApi.sliding-window-size": 10,
        "resilience4j.circuitbreaker.instances.coreApi.minimum-number-of-calls": 5,
        "resilience4j.circuitbreaker.instances.coreApi.failure-rate-threshold": 50,
        "resilience4j.circuitbreaker.instances.coreApi.wait-duration-in-open-state": "20s",
        "resilience4j.circuitbreaker.instances.coreApi.permitted-number-of-calls-in-half-open-state": 3,
        "resilience4j.circuitbreaker.instances.coreApi.automatic-transition-from-open-to-half-open-enabled": false,
        "resilience4j.circuitbreaker.instances.coreApi.register-health-indicator": true,
        "resilience4j.circuitbreaker.instances.coreApi.ignore-exceptions[0]": "com.bank.xyz.bff.ate.error.OperationRechazadaException",
        "resilience4j.retry.retry-aspect-order": 2,
        "resilience4j.retry.instances.coreApi.max-attempts": 3,
        "resilience4j.retry.instances.coreApi.wait-duration": "400ms",
        "resilience4j.retry.instances.coreApi.exponential-backoff-multiplier": 2,
        "resilience4j.retry.instances.coreApi.ignore-exceptions[0]": "com.bank.xyz.bff.ate.error.OperationRechazadaException",
        "resilience4j.timelimiter.instances.coreApi.timeout-duration": "8s"
      }
    },
    {
      "name": "classpath:/config-repo/application.yml",
      "source": {
        "banco.moneda": "CLP",
        "banco.zona-horaria": "America/Santiago",
        "eureka.client.service-url.defaultZone": "${EUREKA_URL:http://localhost:8761/eureka}",
        "eureka.client.register-with-eureka": true,
        "eureka.client.fetch-registry": true,
        "eureka.instance.prefer-ip-address": true,
        "management.endpoints.web.exposure.include": "health,info,metrics,circuitbreakers,circuitbreakerevents,refresh",
        "management.endpoint.health.show-details": "always",
        "logging.pattern.console": "%d{HH:mm:ss.SSS} [%X-14.14thread] %X-51level %X-26.26logger{0} : %msg%n"
      }
    }
  ]
}
```

Servicio de dominio

La API del dominio, unico modulo con acceso a PostgreSQL. El endpoint de retiro esta marcado como operacion critica e idempotente por el campo referencia.

 Swagger
Open source SMARTER

/v3/api-docs

Explore

Banco XYZ - Servicio de dominio

1.0 OA & 3.0

/v3/api-docs

API generica sobre los datos cargados por el batch. La consumen los tres BFF, ningun cliente la usa directamente.

Servers

http://localhost:8080 - Generated server uri

Transacciones

Resumen diario de transacciones

GET

/api/v1/transacciones/resumen-diario

Resumen diario, opcionalmente acotado por rango de fechas

Retiros

Operacion critica: debita el saldo de la cuenta

POST

/api/v1/cuentas/{cuentaId}/retiros

Registra un retiro. Idempotente por el campo referencia

Cuentas

Datos de cuentas generados por el batch de la semana 3

GET

/api/v1/cuentas

Lista paginado de cuentas

GET

/api/v1/cuentas/{cuentaId}

Detalle de una cuenta

GET

/api/v1/cuentas/{cuentaId}/saldo

Saldo actual de la cuenta

GET

/api/v1/cuentas/{cuentaId}/movimientos

Movimientos de la cuenta, opcionalmente filtrados por año

GET

/api/v1/cuentas/{cuentaId}/movimientos/ultimos

Ultimos movimientos, pensado para clientes livianos

BFF de cajeros

Superficie minima: solo saldo y retiro. Los candados indican que ambas operaciones exigen la credencial de terminal, ademas del token OAuth2 que el BFF usa hacia el dominio.

Swagger

/v3/api-docs

Explore

Banco XYZ - BFF Cajeros

1.0

OAuth 3.0

/v3/api-docs

Backend dedicado a los cajeros automaticos. Superficie minima, autentificacion por terminal, validaciones de denominacion y auditoria. Usa el boton Authorize para cargar las cabeceras X-ATM-Terminal y X-ATM-Key.

Servers

http://localhost:8083 - Generated server uri

Authorize

BFF Cajeros

Operaciones críticas. Requiere credencial de terminal

POST

/bff/atm/cuentas/{cuentaId}/retiro

Retiro de efectivo. Idempotente si se repite la referencia

GET

/bff/atm/cuentas/{cuentaId}/saldo

Consulta de saldo con el maximo retirable ya calculado

Schemas

RetiroAtmRequest

ComprobanteRetiro

SaldoAtm

BFF del portal

Respuestas completas para el navegador: el endpoint de panel arma en una sola llamada lo que la pantalla necesita.

 Swagger
Open Source API Documentation

/v3/api-docs

Explore

Banco XYZ - BFF Web

1.0OAS 3.0

/v3/api-docs

Backend dedicado al portal de navegador. Agrega en una sola respuesta los datos que la pantalla necesita.

Servers

http://localhost:8081 - Generated server url

BFF Web

Respuestas completas para el portal de navegador

GET/bff/web/cuentas

Listado paginado de cuentas para la grilla del portal

GET/bff/web/cuentas/{cuentaId}/panel

Vista completa de la cuenta en una sola llamada

Schemas

Cuenta

PaginaCuenta

EstadoAnual

Metadatos

Movimiento

BFF movil

El mismo dominio, pero devolviendo solo lo esencial para reducir el consumo de datos.

 Swagger
Open Source API Documentation

/v3/api-docs

Explore

Banco XYZ - BFF Movil

1.0 OA 3.0

[/v3/api-docs](#)

Backend dedicado a la app movil. Devuelve solo los datos esenciales para reducir consumo de datos y latencia.

Servers

http://localhost:8082 - Generated server url

BFF Movil

Respuestas minimas para la app de telefono

GET

/bff/movil/cuentas/{cuentaId}

Pantalla principal: saldo y ultimos cinco movimientos

GET

/bff/movil/cuentas/{cuentaId}/saldo

Solo el saldo, la consulta mas frecuente de la app

Schemas

MovimientoMovil

ResumenMovil

SaldoMovil

Servicio de notificaciones

El consumidor de eventos. No lo llama nadie: se alimenta del topico JMS.

 **Swagger**
OpenAPI 3.0

/v3/api-docs

Explore

OpenAPI definition

v0 OAS 3.0

/v3/api-docs

Servers

http://localhost:8084 - Generated server url

Notificaciones

Eventos de transaccion consumidos del broker

GET

/api/v1/notificaciones

Ultimas notificaciones generadas

⌵

GET

/api/v1/notificaciones/resumen

Cantidad de eventos procesados

⌵

GET

/api/v1/notificaciones/cuentas/{cuentaId}

Notificaciones de una cuenta

⌵

Schemas

⌵

Notificacion >

Estado del circuit breaker

El actuador exponiendo la instancia coreApi en estado CLOSED, con su umbral de fallos en 50 % y el conteo de llamadas de la ventana.

```
pretty-print ☐  
{"circuitBreakers":{"coreApi":  
{"failureRate":"-1.8%", "slowCallRate":"-1.8%", "failureRateThreshold":"50.8%", "slowCallRateThreshold":"100.8%", "bufferedCalls":1, "failedCalls":0, "slowCalls":0, "slowFailedCalls":0, "notPermittedCalls":0, "state":"CLOSED"}}}
```

Notificacion generada por un evento

La notificacion que produjo el servicio consumidor a partir del evento que publico el dominio tras confirmar el retiro. El eventoid coincide con la referencia del retiro.

pretty-print ☐

```
[{"eventoId":"PRUEBA-203223","cuentaId":101,"titulo":"Retiro realizado","mensaje":"Se registro un retiro realizado por $3.000 en tu cuenta 101. Saldo disponible: $95.814.", "canal":"ATW","recibidoEn":"2026-10-04T20:32:24.7062832"}]
```

3. Salidas de consola

Generadas con el script `evidencias\generar-evidencia-microservicios.ps1`, que consulta los servicios en ejecucion y escribe los resultados sin intervencion manual.

Configuracion centralizada y Service Discovery

```
=====
SEMANA 6 - Configuracion centralizada y Service Discovery
Generado: 2026-10-04 15:57:00
=====

El config server entrega la configuracion a los microservicios y Eureka mantiene
el registro de instancias. Ningun microservicio tiene la configuracion de negocio
en su propio archivo.

--- Salud del config server ---

GET http://localhost:8888/actuator/health

HTTP 200    (0.029335 s)

{"status":"UP"}

--- Configuracion que el config server entrega a banco-core-api ---

GET http://localhost:8888/banco-core-api/default

HTTP 200    (0.040293 s)

{"name":"banco-core-api","profiles":["default"],"label":null,"version":null,"state":null,"propertySources":[{"name":"classpath:/config-repo/banco-core-api.yml","source":{"banco.atm.monto-maximo":200000,"banco.atm.multiplo":1000,"banco.eventos.destino":"banco.eventos.transaccion","spring.datasource.url":"${DB_URL:jdbc:postgresql://localhost:5432/bank_legacy_db}","spring.datasource.username":"${DB_USER:postgres}","spring.datasource.password":"${DB_PASSWORD:postgres}","spring.jpa.hibernate.ddl-auto":"update","spring.jpa.open-in-view":false,"spring.artemis.mode":"${ARTEMIS_MODE:embedded}","spring.artemis.broker-url":"${ARTEMIS_URL:tcp://localhost:61616}","spring.artemis.embedded.enabled":"${ARTEMIS_EMBEDDED:true}","spring.artemis.embedded.persistent":false,"spring.jms.pub-sub-domain":true,"spring.security.oauth2.resourceserver.jwt.issuer-uri":"${AUTH_URL:http://localhost:9000}"}}, {"name":"classpath:/config-repo/application.yml","source":{"banco.moneda":"CLP","banco.zona-horaria":"America/Santiago","eureka.client.service-url.defaultZone":"${EUREKA_URL:http://localhost:8761/eureka/}","eureka.client.register-with-eureka":true,"eureka.client.fetch-registry":true,"eureka.instance.prefer-ip-address":true,"management.endpoints.web.exposure.include":"health,info,metrics,circuitbreakers,circuitbreakerevents,refresh","management.endpoint.health.show-details":"always","logging.pattern.console":"%d{HH:mm:ss.SSS} [%-14.14thread] %-5level %-26.26logger{0} : %msg%n"}}]}

NOTA: Las propiedades salen de config-repo/banco-core-api.yml y de application.yml, no del jar del microservicio.

--- Configuracion de resiliencia que recibe bff-atm ---

GET http://localhost:8888/bff-atm/default

HTTP 200    (0.016412 s)

{"name":"bff-atm","profiles":["default"],"label":null,"version":null,"state":null,"propertySources":[{"name":"classpath:/config-repo/bff-atm.yml","source":{"banco.core-api.url":"${CORE_API_URL:http://localhost:8080}","banco.atm.monto-minimo":1000,"banco.atm.monto-maximo":200000,"banco.atm.denominacion":1000,"resilience4j.circuitbreaker.circuit-breaker-aspect-order":3,"resilience4j.circuitbreaker.instances.coreApi.sliding-window-type":"COUNT_BASED","resilience4j.circuitbreaker.instances.coreApi.sliding-window-size":10,"resilience4j.circuitbreaker.instances.coreApi.minimum-number-of-calls":5,"resilience4j.circuitbreaker.instances.coreApi.failure-rate-threshold":50,"resilience4j.circuitbreaker.instances.coreApi.wait-duration-in-open-state":"20s","resilience4j.circuitbreaker.instances.coreApi.permitted-number-of-calls-in-half-open-state":3,"resilience4j.circuitbreaker.instances.coreApi.automatic-transition-from-open-to-half-open-enabled":false,"resilience4j.circuitbreaker.instances.coreApi.register-health-indicator":true,"resilience4j.circuitbreaker.instances.coreApi.ignore-exceptions[0]":"com.bank.xyz.bff.atm.error OperacionRechazadaException","resilience4j.retry.retry-aspect-order":2,"resilience4j.retry.instances.coreApi.max-attempts":3,"resilience4j.retry.instances.coreApi.wait-duration":"400ms","resilience4j.retry.instances.coreApi.exponential-backoff-multiplier":2,"resilience4j.retry.instances.coreApi.ignore-exceptions[0]":"com.bank.xyz.bff.atm.error OperacionRechazadaException","resilience4j.timelimiter.instances.coreApi.timeout-duration":"8s"}}, {"name":"classpath:/config-repo/application.yml","source":{"banco.moneda":"CLP","banco.zona-horaria":"America/Santiago","eureka.client.service-url.defaultZone":"${EUREKA_URL:http://localhost:8761/eureka/}","eureka.client.register-with-eureka":true,"eureka.client.fetch-registry":true,"eureka.instance.prefer-ip-address":true,"management.endpoints.web.exposure.include":"health,info,metrics,circuitbreakers,circuitbreakerevents,refresh","management.endpoint.health.show-details":"always","logging.pattern.console":"%d{HH:mm:ss.SSS} [%-14.14thread] %-5level %-26.26logger{0} : %msg%n"}}]}

NOTA: Los umbrales del circuit breaker se administran de forma centralizada.

--- Registro de Eureka ---

GET http://localhost:8761/eureka/apps
Accept: application/json

HTTP 200    (0.002542 s)

{"applications":{"versions__delta":"1","apps__hashcode":"UP_5_","application":[{"name":"BFF-MOBILE","instance":{"instanceId":"LAPTOP-2RGGN8RD:bff-mobile:8082","hostName":"192.168.68.100","app":"BFF-MOBILE","ipAddr":"192.168.68.100","status":"UP","overriddenStatus":"UNKNOWN","port":{"$":8082,"@enabled":"true"},"securePort":
```

```

{"$":443,"@enabled":"false"},"countryId":1,"dataCenterInfo":
{"@class":"com.netflix.appinfo.InstanceInfo$DefaultDataCenterInfo","name":"MyOwn"},"leaseInfo":
{"renewalIntervalInSecs":30,"durationInSecs":90,"registrationTimestamp":1791143811544,"lastRenewalTimestamp":1791143811544,"evictionTimesta
mp":0,"serviceUpTimestamp":1791142659279},"metadata":
{"management.port":"8082"},"homePageUrl":"http://192.168.68.100:8082/","statusPageUrl":"http://192.168.68.100:8082/actuator/info","healthCh
eckUrl":"http://192.168.68.100:8082/actuator/health","vipAddress":"bff-mobile","secureVipAddress":"bff-
mobile","isCoordinatingDiscoveryServer":"false","lastUpdatedTimestamp":1791143811544,"lastDirtyTimestamp":1791143810752,"actionType":"A
DDED"}},{ "name":"BFF-ATM","instance":[{"instanceId":"LAPTOP-2RGGN8RD:bff-atm:8083","hostName":"192.168.68.100","app":"BFF-
ATM","ipAddr":"192.168.68.100","status":"UP","overriddenStatus":"UNKNOWN","port":{"$:8083,"@enabled":"true"},"securePort":
{"$:443,"@enabled":"false"},"countryId":1,"dataCenterInfo":
{"@class":"com.netflix.appinfo.InstanceInfo$DefaultDataCenterInfo","name":"MyOwn"},"leaseInfo":
{"renewalIntervalInSecs":30,"durationInSecs":90,"registrationTimestamp":1791143519989,"lastRenewalTimestamp":1791143910164,"evictionTimesta
mp":0,"serviceUpTimestamp":1791143213027},"metadata":
{"management.port":"8083"},"homePageUrl":"http://192.168.68.100:8083/","statusPageUrl":"http://192.168.68.100:8083/actuator/info","healthCh
eckUrl":"http://192.168.68.100:8083/actuator/health","vipAddress":"bff-atm","secureVipAddress":"bff-
atm","isCoordinatingDiscoveryServer":"false","lastUpdatedTimestamp":1791143519989,"lastDirtyTimestamp":1791143519400,"actionType":"ADDE
D"}},{ "name":"SERVICIO-NOTIFICACIONES","instance":[{"instanceId":"LAPTOP-2RGGN8RD:servicio-
notificaciones:8084","hostName":"192.168.68.100","app":"SERVICIO-
NOTIFICACIONES","ipAddr":"192.168.68.100","status":"UP","overriddenStatus":"UNKNOWN","port":{"$:8084,"@enabled":"true"},"securePort":
{"$:443,"@enabled":"false"},"countryId":1,"dataCenterInfo":
{"@class":"com.netflix.appinfo.InstanceInfo$DefaultDataCenterInfo","name":"MyOwn"},"leaseInfo":
{"renewalIntervalInSecs":30,"durationInSecs":90,"registrationTimestamp":1791142807296,"lastRenewalTimestamp":1791143888131,"evictionTimesta
mp":0,"serviceUpTimestamp":1791142658540},"metadata":
{"management.port":"8084"},"homePageUrl":"http://192.168.68.100:8084/","statusPageUrl":"http://192.168.68.100:8084/actuator/info","healthCh
eckUrl":"http://192.168.68.100:8084/actuator/health","vipAddress":"servicio-notificaciones","secureVipAddress":"servicio-
notificaciones","isCoordinatingDiscoveryServer":"false","lastUpdatedTimestamp":1791142807296,"lastDirtyTimestamp":1791142806725,"action
Type":"ADDED"}},{ "name":"BANCO-CORE-API","instance":[{"instanceId":"LAPTOP-2RGGN8RD:banco-core-
api:8080","hostName":"192.168.68.100","app":"BANCO-CORE-API","ipAddr":"192.168.68.100","status":"UP","overriddenStatus":"UNKNOWN","port":
{"$:8080,"@enabled":"true"},"securePort":{"$:443,"@enabled":"false"},"countryId":1,"dataCenterInfo":
{"@class":"com.netflix.appinfo.InstanceInfo$DefaultDataCenterInfo","name":"MyOwn"},"leaseInfo":
{"renewalIntervalInSecs":30,"durationInSecs":90,"registrationTimestamp":1791143564280,"lastRenewalTimestamp":1791143894454,"evictionTimesta
mp":0,"serviceUpTimestamp":1791142614717},"metadata":
{"management.port":"8080"},"homePageUrl":"http://192.168.68.100:8080/","statusPageUrl":"http://192.168.68.100:8080/actuator/info","healthCh
eckUrl":"http://192.168.68.100:8080/actuator/health","vipAddress":"banco-core-api","secureVipAddress":"banco-core-
api","isCoordinatingDiscoveryServer":"false","lastUpdatedTimestamp":1791143564280,"lastDirtyTimestamp":1791143563672,"actionType":"ADDE
D"}},{ "name":"BFF-WEB","instance":[{"instanceId":"LAPTOP-2RGGN8RD:bff-web:8081","hostName":"192.168.68.100","app":"BFF-
WEB","ipAddr":"192.168.68.100","status":"UP","overriddenStatus":"UNKNOWN","port":{"$:8081,"@enabled":"true"},"securePort":
{"$:443,"@enabled":"false"},"countryId":1,"dataCenterInfo":
{"@class":"com.netflix.appinfo.InstanceInfo$DefaultDataCenterInfo","name":"MyOwn"},"leaseInfo":
{"renewalIntervalInSecs":30,"durationInSecs":90,"registrationTimestamp":1791143811545,"lastRenewalTimestamp":1791143811545,"evictionTimesta
mp":0,"serviceUpTimestamp":1791142659824},"metadata":
{"management.port":"8081"},"homePageUrl":"http://192.168.68.100:8081/","statusPageUrl":"http://192.168.68.100:8081/actuator/info","healthCh
eckUrl":"http://192.168.68.100:8081/actuator/health","vipAddress":"bff-web","secureVipAddress":"bff-
web","isCoordinatingDiscoveryServer":"false","lastUpdatedTimestamp":1791143811545,"lastDirtyTimestamp":1791143810955,"actionType":"ADDE
D"}]]}}

```

NOTA: Los cinco microservicios estan registrados y en estado UP.

```

=====
SEMANA 8 - Seguridad con OAuth2
Generado: 2026-10-04 15:57:00
=====

El servidor de autorizacion emite tokens JWT con el flujo client_credentials.
banco-core-api actua como resource server y autoriza por scope.

--- Metadatos del servidor de autorizacion ---

GET http://localhost:9000/.well-known/oauth-authorization-server

HTTP 200 (0.029233 s)

{"issuer":"http://localhost:9000","authorization_endpoint":"http://localhost:9000/oauth2/authorize","device_authorization_endpoint":"http://localhost:9000/oauth2/device_authorization","token_endpoint":"http://localhost:9000/oauth2/token","token_endpoint_auth_methods_supported":["client_secret_basic","client_secret_post","client_secret_jwt","private_key_jwt"],"jwks_uri":"http://localhost:9000/oauth2/jwks","response_types_supported":["code"],"grant_types_supported":["authorization_code","client_credentials","refresh_token","urn:ietf:params:oauth:grant-type:device_code"],"revocation_endpoint":"http://localhost:9000/oauth2/revoke","revocation_endpoint_auth_methods_supported":["client_secret_basic","client_secret_post","client_secret_jwt","private_key_jwt"],"introspection_endpoint":"http://localhost:9000/oauth2/introspect","introspection_endpoint_auth_methods_supported":["client_secret_basic","client_secret_post","client_secret_jwt","private_key_jwt"],"code_challenge_methods_supported":["S256"]}

--- Token para bff-atm (client_credentials) ---

POST http://localhost:9000/oauth2/token
-u bff-atm:secreto-bff-atm
grant_type=client_credentials&scope=cuentas.leer retiros.escribir

HTTP 200 (0.090798 s)

{"access_token":"eyJraWQwIjIzjY50TQwSY0ZnN2U5LTQ30TUtYWMzOS1jNzAwZTgzZWUxZmQlLCJhbGciOiJSUzI1NiJ9.eyJzdWIiOiIjZmYtYXRtIiwiaXVkJoiYmZmLWF0bSIsIm5iZiI6MTc5MTE0MzgyMSwic2NvcGU0IjIzY3VlbnRlcy5lcnNyYWJpc1IsImN1ZS05YXMuYmVlciJldLCjpc3MiOiJodHRwOi8vbG9jYXRob3N0OjkwMDAiLCJleHAiOiE30TEyNDU2MjEsImhhdCI6MTc5MTE0MzgyMSwianRpIjoimZBmYjYzMDYtMGU4ZS00OTk5LWE5NGQtMGU2NGQyNzdjYTliIn0.LLNiEgJam7PjWwOdw-icWcsvbdLGuIWKjRWNSpbJPbv6XyRsCUCMraXkz-X0buTn3VzE3Z9MCN2fQPJhHFU_cQHwQc1E4uSdVYomPwY6FzNPNwYf90oSv0ym3KadMwZ8tIVZEiuZ306BmilySS2-wQz-yywv026U1RNag7DF0tNYGeTxZ6en_dBFKcPu1Hy7y-8ixnDpbcdnhQSeYu_MKq_gmhAJpWerTtKYsxMyKIEVCnNjGwk0jrUsbzXwZHVn0_FqZmjDNsYNMbzbjz991T-gB6gTF_yuEIIPK78UoNo2J1tWJXquTXyQQHLY_uydMYntTX7F4Y_t4T-4Q111A","scope":"retiros.escribir cuentas.leer","token_type":"Bearer","expires_in":1800}

NOTA: El cajero obtiene scopes de lectura y de escritura de retiros.

--- Token para bff-mobile (scope reducido) ---

POST http://localhost:9000/oauth2/token
-u bff-mobile:secreto-bff-mobile
grant_type=client_credentials&scope=cuentas.leer

HTTP 200 (0.091463 s)

{"access_token":"eyJraWQwIjIzjY50TQwSY0ZnN2U5LTQ30TUtYWMzOS1jNzAwZTgzZWUxZmQlLCJhbGciOiJSUzI1NiJ9.eyJzdWIiOiIjZmYtYTBwIiwiaXVkJoiYmZmLWl1YmIsZSIsIm5iZiI6MTc5MTE0MzgyMSwic2NvcGU0IjIzY3VlbnRlcy5lcnNyYWJpc1IsImN1ZS05YXMuYmVlciJldLCjpc3MiOiJodHRwOi8vbG9jYXRob3N0OjkwMDAiLCJleHAiOiE30TEyNDU2MjEsImhhdCI6MTc5MTE0MzgyMSwianRpIjoimZBmYjYzMDYtMGU4ZS00OTk5LWE5NGQtMGU2NGQyNzdjYTliIn0.LLNiEgJam7PjWwOdw-icWcsvbdLGuIWKjRWNSpbJPbv6XyRsCUCMraXkz-6In7GMkrV607W","scope":"cuentas.leer","token_type":"Bearer","expires_in":1800}

NOTA: La app movil solo puede leer: no tiene el scope retiros.escribir.

--- Acceso al recurso protegido SIN token ---

GET http://localhost:8080/api/v1/cuentas/101

HTTP 401 (0.006947 s)

NOTA: Rechazado con 401 por el resource server.

--- Acceso al recurso protegido CON token ---

GET http://localhost:8080/api/v1/cuentas/101
Authorization: Bearer
eyJraWQwIjIzjY50TQwSY0ZnN2U5LTQ30TUtYWMzOS1jNzAwZTgzZWUxZmQlLCJhbGciOiJSUzI1NiJ9.eyJzdWIiOiIjZmYtYXRtIiwiaXVkJoiYmZmLWF0bSIsIm5iZiI6MTc5MTE0MzgyMSwic2NvcGU0IjIzY3VlbnRlcy5lcnNyYWJpc1IsImN1ZS05YXMuYmVlciJldLCjpc3MiOiJodHRwOi8vbG9jYXRob3N0OjkwMDAiLCJleHAiOiE30TEyNDU2MjEsImhhdCI6MTc5MTE0MzgyMSwianRpIjoimZBmYjYzMDYtMGU4ZS00OTk5LWE5NGQtMGU2NGQyNzdjYTliIn0.LLNiEgJam7PjWwOdw-icWcsvbdLGuIWKjRWNSpbJPbv6XyRsCUCMraXkz-
```

```
X0buTn3vZe3Z9MCN2fQPJhHFU_cQHwqC1E4uSdVYomPwY6FzNPNwYf90oSV0ym3KadMwZ8tIVZEiuZ306Bmi1ySS2-wQz-yywvebO26U1RNag7DF0tNYGeTxZ6en_dBFKcPu1Hy7y-8ixnDpbcdnhQSeYu_MKq_gmhAJPwEr-TtKYsxMyKIeVCnNjGwk0jUrUsbzXwZHVnO_FqZmjDNsYNMbZjz991T-gBGgTF_yuEIIIPK78UoNo2J1tWJXquTXyQQHLY_uYdmYntTX7F4Y_t4T-4Q111A
```

```
HTTP 200 (0.013339 s)
```

```
{"cuentaId":101,"nombre":"Bob Johnson","tipo":"hipoteca","saldoFinal":103814.17,"interesTotal":814.17,"registrosProcesados":13,"actualizadoEn":"2026-10-04T15:40:20.100734"}
```

```
--- Credenciales de cliente incorrectas ---
```

```
POST http://localhost:9000/oauth2/token
-u bff-atm:clave-equivocada
grant_type=client_credentials
```

```
HTTP 401 (0.068862 s)
```

```
{"error":"invalid_client"}
```

Arquitectura de eventos: publicacion, consumo e idempotencia

```
=====
SEMANA 7 - Arquitectura de eventos con JMS
Generado: 2026-10-04 15:57:01
=====

Un retiro confirmado publica un evento en el topico banco.eventos.transaccion.
El servicio de notificaciones lo consume de forma asincrona y genera el aviso.

--- Notificaciones antes del retiro ---

GET http://localhost:8084/api/v1/notificaciones/resumen

HTTP 200    (0.020409 s)

{"totalNotificaciones":2}

--- Retiro desde el cajero ---

POST http://localhost:8083/bff/atm/cuentas/101/retiro
X-ATM-Terminal: ATM-001
X-ATM-Key: clave-demo-001
body: {"monto":5000,"referencia":"EVID-20261004-155701"}

HTTP 200    (0.439465 s)

{"comprobante":"EVID-20261004-155701","cuenta":101,"montoEntregado":5000,"saldoResultante":98814.17,"terminal":"ATM-001","duplicado":false,"fechaHora":"2026-10-04T15:57:01.718045"}

NOTA: bff-atm pide el token, llama al dominio y este publica el evento tras confirmar el debito.

--- Notificacion generada por el consumidor ---

GET http://localhost:8084/api/v1/notificaciones?cantidad=3

HTTP 200    (0.013225 s)

[{"eventoId":"EVID-20261004-155701","cuentaId":101,"titulo":"Retiro realizado","mensaje":"Se registro un retiro realizado por $5.000 en tu cuenta 101. Saldo disponible: $98.814.", "canal":"ATM", "recibidoEn":"2026-10-04T15:57:01.8263679"}, {"eventoId":"IDEM-154039","cuentaId":102,"titulo":"Retiro realizado","mensaje":"Se registro un retiro realizado por $3.000 en tu cuenta 102. Saldo disponible: $115.812.", "canal":"ATM", "recibidoEn":"2026-10-04T15:40:39.4394913"}, {"eventoId":"EVT-154019","cuentaId":101,"titulo":"Retiro realizado","mensaje":"Se registro un retiro realizado por $5.000 en tu cuenta 101. Saldo disponible: $103.814.", "canal":"ATM", "recibidoEn":"2026-10-04T15:40:20.2566465"}]

NOTA: El servicio de notificaciones nunca fue invocado directamente: reacciono al evento.

--- Mismo retiro repetido (idempotencia) ---

POST http://localhost:8083/bff/atm/cuentas/101/retiro
X-ATM-Terminal: ATM-001
X-ATM-Key: clave-demo-001
body: {"monto":5000,"referencia":"EVID-20261004-155701"}

HTTP 200    (0.042290 s)

{"comprobante":"EVID-20261004-155701","cuenta":101,"montoEntregado":5000.00,"saldoResultante":98814.17,"terminal":"ATM-001","duplicado":true,"fechaHora":"2026-10-04T15:57:01.718045"}

NOTA: duplicado=true y el saldo no vuelve a bajar.

--- El consumidor no duplico el aviso ---

GET http://localhost:8084/api/v1/notificaciones/resumen

HTTP 200    (0.020553 s)

{"totalNotificaciones":3}

NOTA: El evento repetido se descarta por su eventoId.
```

Tolerancia a fallos: ciclo completo del circuit breaker

```
=====
SEMANA 7 - Tolerancia a fallos con Resilience4j
Generado: 2026-10-04 15:57:08
=====

bff-atm protege sus llamadas al dominio con circuit breaker, reintentos y
limite de tiempo. El estado del circuito se observa por actuador.

--- Estado del circuito en operacion normal ---

GET http://localhost:8083/actuador/circuitbreakers

HTTP 200   (0.013762 s)

{"circuitBreakers":{"coreApi":
{"failureRate":"-1.0%","slowCallRate":"-1.0%","failureRateThreshold":"50.0%","slowCallRateThreshold":"100.0%","bufferedCalls":4,"failedCall
s":0,"slowCalls":0,"slowFailedCalls":0,"notPermittedCalls":0,"state":"CLOSED"}}}

NOTA: CLOSED: las llamadas pasan al servicio de dominio.

--- Consulta de saldo correcta ---

GET http://localhost:8083/bff/atm/cuentas/101/saldo
X-ATM-Terminal: ATM-001
X-ATM-Key: clave-demo-001

HTTP 200   (0.033880 s)

{"cuenta":101,"saldoDisponible":98814.17,"maximoRetirable":98000.00,"denominacion":1000,"consultadoEn":"2026-10-04T15:57:08.1793453"}

--- Ciclo completo del circuit breaker ---

Para reproducirlo hay que detener banco-core-api, hacer seis consultas
de saldo, volver a levantarlo, esperar los 20 s de wait-duration y
consultar de nuevo. Transiciones observadas en la ejecucion de prueba:

CLOSED_TO_OPEN      tras 5 llamadas fallidas (umbral 50 %)
OPEN_TO_HALF_OPEN   al reintentar despues de la espera
HALF_OPEN_TO_CLOSED tras 3 llamadas correctas

Con el circuito abierto el cajero recibe:
HTTP 503 {"codigo":"SIN_SERVICIO","entregarEfectivo":false}
y no entrega dinero.

--- Eventos registrados del circuito ---

GET http://localhost:8083/actuador/circuitbreakerevents/coreApi

HTTP 200   (0.004197 s)

{"circuitBreakerEvents":[{"circuitBreakerName":"coreApi","type":"ERROR","creationTime":"2026-10-04T15:52:21.670470800-
04:00[America/La_Paz]","errorMessage":"com.bank.xyz.bff.atm.error.CoreApiNoDisponibleException: sin conexion con el servicio de
dominio","durationInMs":108,"stateTransition":null},{"circuitBreakerName":"coreApi","type":"ERROR","creationTime":"2026-10-
04T15:52:22.097531200-04:00[America/La_Paz]","errorMessage":"com.bank.xyz.bff.atm.error.CoreApiNoDisponibleException: sin conexion con el
servicio de dominio","durationInMs":14,"stateTransition":null},{"circuitBreakerName":"coreApi","type":"ERROR","creationTime":"2026-10-
04T15:52:22.515013700-04:00[America/La_Paz]","errorMessage":"com.bank.xyz.bff.atm.error.CoreApiNoDisponibleException: sin conexion con el
servicio de dominio","durationInMs":15,"stateTransition":null},{"circuitBreakerName":"coreApi","type":"ERROR","creationTime":"2026-10-
04T15:52:22.575439300-04:00[America/La_Paz]","errorMessage":"com.bank.xyz.bff.atm.error.CoreApiNoDisponibleException: sin conexion con el
servicio de dominio","durationInMs":0,"stateTransition":null},{"circuitBreakerName":"coreApi","type":"ERROR","creationTime":"2026-10-
04T15:52:23.005286500-04:00[America/La_Paz]","errorMessage":"com.bank.xyz.bff.atm.error.CoreApiNoDisponibleException: sin conexion con el
servicio de dominio","durationInMs":15,"stateTransition":null},
{"circuitBreakerName":"coreApi","type":"FAILURE_RATE_EXCEEDED","creationTime":"2026-10-04T15:52:23.005854400-
04:00[America/La_Paz]","errorMessage":null,"durationInMs":null,"stateTransition":null},
{"circuitBreakerName":"coreApi","type":"STATE_TRANSITION","creationTime":"2026-10-04T15:52:23.008436600-
04:00[America/La_Paz]","errorMessage":null,"durationInMs":null,"stateTransition":"CLOSED_TO_OPEN"},
{"circuitBreakerName":"coreApi","type":"NOT_PERMITTED","creationTime":"2026-10-04T15:52:23.411252-
04:00[America/La_Paz]","errorMessage":null,"durationInMs":null,"stateTransition":null},
{"circuitBreakerName":"coreApi","type":"NOT_PERMITTED","creationTime":"2026-10-04T15:52:23.475395400-
04:00[America/La_Paz]","errorMessage":null,"durationInMs":null,"stateTransition":null},
{"circuitBreakerName":"coreApi","type":"NOT_PERMITTED","creationTime":"2026-10-04T15:52:23.884835-
04:00[America/La_Paz]","errorMessage":null,"durationInMs":null,"stateTransition":null},
{"circuitBreakerName":"coreApi","type":"NOT_PERMITTED","creationTime":"2026-10-04T15:52:24.291397900-
04:00[America/La_Paz]","errorMessage":null,"durationInMs":null,"stateTransition":null},
{"circuitBreakerName":"coreApi","type":"NOT_PERMITTED","creationTime":"2026-10-04T15:52:24.357960800-
04:00[America/La_Paz]","errorMessage":null,"durationInMs":null,"stateTransition":null},
{"circuitBreakerName":"coreApi","type":"NOT_PERMITTED","creationTime":"2026-10-04T15:52:24.768485800-
04:00[America/La_Paz]","errorMessage":null,"durationInMs":null,"stateTransition":null},
{"circuitBreakerName":"coreApi","type":"NOT_PERMITTED","creationTime":"2026-10-04T15:52:25.173822100-
```

```
04:00[America/La_Paz]", "errorMessage": null, "durationInMs": null, "stateTransition": null},
{"circuitBreakerName": "coreApi", "type": "NOT_PERMITTED", "creationTime": "2026-10-04T15:52:25.238546100-04:00[America/La_Paz]", "errorMessage": null, "durationInMs": null, "stateTransition": null},
{"circuitBreakerName": "coreApi", "type": "NOT_PERMITTED", "creationTime": "2026-10-04T15:52:25.653658600-04:00[America/La_Paz]", "errorMessage": null, "durationInMs": null, "stateTransition": null},
{"circuitBreakerName": "coreApi", "type": "NOT_PERMITTED", "creationTime": "2026-10-04T15:52:26.058248900-04:00[America/La_Paz]", "errorMessage": null, "durationInMs": null, "stateTransition": null},
{"circuitBreakerName": "coreApi", "type": "NOT_PERMITTED", "creationTime": "2026-10-04T15:52:26.125442500-04:00[America/La_Paz]", "errorMessage": null, "durationInMs": null, "stateTransition": null},
{"circuitBreakerName": "coreApi", "type": "NOT_PERMITTED", "creationTime": "2026-10-04T15:52:26.529004600-04:00[America/La_Paz]", "errorMessage": null, "durationInMs": null, "stateTransition": null},
{"circuitBreakerName": "coreApi", "type": "NOT_PERMITTED", "creationTime": "2026-10-04T15:52:26.945946400-04:00[America/La_Paz]", "errorMessage": null, "durationInMs": null, "stateTransition": null},
{"circuitBreakerName": "coreApi", "type": "STATE_TRANSITION", "creationTime": "2026-10-04T15:53:06.453484400-04:00[America/La_Paz]", "errorMessage": null, "durationInMs": null, "stateTransition": "OPEN_TO_HALF_OPEN"},
{"circuitBreakerName": "coreApi", "type": "SUCCESS", "creationTime": "2026-10-04T15:53:06.728251100-04:00[America/La_Paz]", "errorMessage": null, "durationInMs": 274, "stateTransition": null},
{"circuitBreakerName": "coreApi", "type": "SUCCESS", "creationTime": "2026-10-04T15:53:06.823532200-04:00[America/La_Paz]", "errorMessage": null, "durationInMs": 13, "stateTransition": null},
{"circuitBreakerName": "coreApi", "type": "SUCCESS", "creationTime": "2026-10-04T15:53:06.890938800-04:00[America/La_Paz]", "errorMessage": null, "durationInMs": 12, "stateTransition": null},
{"circuitBreakerName": "coreApi", "type": "STATE_TRANSITION", "creationTime": "2026-10-04T15:53:06.890938800-04:00[America/La_Paz]", "errorMessage": null, "durationInMs": null, "stateTransition": "HALF_OPEN_TO_CLOSED"},
{"circuitBreakerName": "coreApi", "type": "SUCCESS", "creationTime": "2026-10-04T15:53:06.958212400-04:00[America/La_Paz]", "errorMessage": null, "durationInMs": 9, "stateTransition": null},
{"circuitBreakerName": "coreApi", "type": "SUCCESS", "creationTime": "2026-10-04T15:53:07.035240500-04:00[America/La_Paz]", "errorMessage": null, "durationInMs": 10, "stateTransition": null},
{"circuitBreakerName": "coreApi", "type": "SUCCESS", "creationTime": "2026-10-04T15:57:01.882109600-04:00[America/La_Paz]", "errorMessage": null, "durationInMs": 343, "stateTransition": null},
{"circuitBreakerName": "coreApi", "type": "SUCCESS", "creationTime": "2026-10-04T15:57:05.023425200-04:00[America/La_Paz]", "errorMessage": null, "durationInMs": 22, "stateTransition": null},
{"circuitBreakerName": "coreApi", "type": "SUCCESS", "creationTime": "2026-10-04T15:57:08.179345300-04:00[America/La_Paz]", "errorMessage": null, "durationInMs": 7, "stateTransition": null}}]
```

Los tres BFF operando sobre la plataforma

```
=====
Los tres BFF sobre la plataforma de microservicios
Generado: 2026-10-04 15:57:08
=====

Cada BFF obtiene su propio token con los scopes que le corresponden y consume
el servicio de dominio a través de la infraestructura de Spring Cloud.

--- bff-web: panel completo ---

GET http://localhost:8081/bff/web/cuentas/101/panel?tamano=2

HTTP 200    (0.311036 s)

{"titular":{"cuentaId":101,"nombre":"Bob Johnson","tipoCuenta":"hipoteca","identificado":true},"resumen":
{"saldoActual":98814.17,"interesAcumulado":814.17,"registrosProcesados":13,"actualizadoEn":"2026-10-04T15:57:01.718045"},"estadosAnuales":
[{"anio":2024,"cantidadMovimientos":44,"totalDepositos":38000.00,"totalRetiros":10800.00,"totalCompras":12700.00,"totalPagos":1600.00,"saldoNeto":12900.00,"movimientosConAnomalia":1}],{"intereses":
{"tipoCuenta":"hipoteca","tasaAnual":2.50,"registros":13,"interesMensualTotal":814.17},"movimientos":{"items":[{"id":3050,"fecha":"2024-12-
26","tipo":"deposito","monto":100.00,"descripcion":"SIN DESCRIPCION","estado":"CORREGIDO","observaciones":"monto negativo normalizado
(origen: -100); descripcion ausente"},{"id":3457,"fecha":"2024-12-22","tipo":"deposito","monto":2500.00,"descripcion":"Ingreso
navideño","estado":"VALIDO","observaciones":null}],{"pagina":0,"tamano":2,"total":44,"totalPaginas":22,"ultima":false},"meta":
{"generadoEn":"2026-10-04T15:57:08.5012888","milisegundos":247,"avisos":["Hay 1 movimiento(s) marcados como anomalia para revision."]}}

--- bff-mobile: respuesta mínima ---

GET http://localhost:8082/bff/movil/cuentas/101

HTTP 200    (0.180261 s)

{"id":101,"titular":"Bob Johnson","tipo":"hipoteca","saldo":98814.17,"ultimos":[{"f":"2024-12-26","t":"deposito","m":100.00,"d":"SIN
DESCRIPCION"}, {"f":"2024-12-22","t":"deposito","m":2500.00,"d":"Ingreso navideño"}, {"f":"2024-11-24","t":"deposito","m":100.00,"d":"SIN
DESCRIPCION"}, {"f":"2024-11-23","t":"compra","m":1500.00,"d":"Compra en tienda"}, {"f":"2024-11-22","t":"retiro","m":1000.00,"d":"SIN
DESCRIPCION"}]}

--- bff-atm: saldo con máximo retirable ---

GET http://localhost:8083/bff/atm/cuentas/101/saldo
X-ATM-Terminal: ATM-001
X-ATM-Key: clave-demo-001

HTTP 200    (0.022808 s)

{"cuenta":101,"saldoDisponible":98814.17,"maximoRetirable":98000.00,"denominacion":1000,"consultadoEn":"2026-10-04T15:57:08.8290296"}

--- bff-atm sin credencial de terminal ---

GET http://localhost:8083/bff/atm/cuentas/101/saldo

HTTP 401    (0.006587 s)

{"codigo":"NO_AUTORIZADO","mensaje":"terminal no autorizado"}
```

Dockerizacion y orquestacion

```
=====
ORQUESTACION CON DOCKER COMPOSE
Generado: 2026-10-04 16:11:55
=====

> docker compose version
Docker Compose version v5.1.0

-----
VALIDACION DEL ARCHIVO
-----
> docker compose config --quiet
(sin salida: el archivo es valido)

> docker compose config --services
config-server
postgres
auth-server
eureka-server
artemis
servicio-notificaciones
banco-core-api
bff-atm
bff-mobile
bff-web

> docker compose config --images
banco-xyz-bff-atm
banco-xyz-auth-server
banco-xyz-banco-core-api
banco-xyz-bff-web
banco-xyz-config-server
banco-xyz-servicio-notificaciones
banco-xyz-eureka-server
apache/activemq-artemis:2.31.2
postgres:16-alpine
banco-xyz-bff-mobile

-----
CONFIGURACION RESUELTA
-----
> docker compose config
name: banco-xyz
services:
  artemis:
    container_name: banco-artemis
    environment:
      ANONYMOUS_LOGIN: "true"
      ARTEMIS_PASSWORD: artemis
      ARTEMIS_USER: artemis
    healthcheck:
      test:
        - CMD-SHELL
        - curl -sf http://localhost:8161 >/dev/null || exit 1
      timeout: 5s
      interval: 10s
      retries: 12
    image: apache/activemq-artemis:2.31.2
    networks:
      banco: null
    ports:
      - mode: ingress
        target: 61616
        published: "61616"
        protocol: tcp
      - mode: ingress
        target: 8161
        published: "8161"
        protocol: tcp
  auth-server:
```

[...] La salida completa esta en el archivo 06-docker-compose.txt

Sobre el levantamiento de contenedores. El archivo docker-compose.yml quedo validado y con su configuracion resuelta, pero no se pudo registrar un `docker compose up` porque Docker Desktop no publica su pipe de motor en este equipo y el demonio no acepta conexiones. La plataforma completa si quedo ejecutada y registrada sin contenedores, como muestran las capturas y las salidas anteriores.